COMPLIMENTARY FINDINGS REPORT

On-site search relevance

An evidence-first read on Target's on-site search across 110 terms in five query types — each graded 0–3 for relevance by a local LLM judge. The finding: a strong engine, with a clear gap in compositional natural-language intent and a narrower regional-synonym gap.

Prepared for
Target

Prepared by Northvalley Intelligence LLC

A quick, browser-based research read — not a formal audit. No crawling and no load on production; each term run once on the shopper-facing organic search (sponsored excluded) at a human pace, with bot protection respected and never bypassed. Surprising zeros were re-checked live — brand-name queries redirect to curated brand pages and are counted as answered.



THE VERDICT

Target search is strong on product-type, attribute and brand queries and most synonyms (overall grade 1.76/3 across 110 terms). Its weakness is compositional natural-language intent — phrases like “move into a new house” that do not map to a single category — plus a narrower regional-synonym gap (“pop”, “popsicle”).

MEAN GRADE

1.76/3

38% of 721 graded results not-relevant.

Summary

THE HEADLINE — SEARCH FIXES AT SCALE

Across 83 demand-weighted search terms, poor relevance puts an estimated **$275K/year** at risk. Applied to the **top 1,000** search terms, that projects to **$3.3M per year** (range $853K–$8.3M) — the prize for fixing search. Modeled on real Google demand; assumptions and full math in the appendix.

STRONG, WITH FOCUSED GAPS

Target search handles the everyday well — synonyms, brand and product-name queries, and product types all return relevant results (overall grade 1.76/3 across 110 terms). The clear weakness is compositional, natural-language intent (job-function queries grade just 1.33/3): when a phrase does not map to a single category, the engine keyword-matches a token and returns a plausible aisle rather than understanding the request.

The intent gap is specific: "move into a new house" returns romance novels and toy food trucks; "set up a home theater" mixes soundbars with women's pajamas; "furnish a living room" returns vases (not furniture). But it handles intent well when the phrase maps to a real category — "healthy snacks for kids", "home gym equipment", "organize my kitchen" all work. A second, narrower gap is regional/substring synonyms: the engine substring-matches rather than mapping the word, so "pop" and "popsicle" return Pop-Tarts, "ice lolly" returns mints, and "jumper" returns baby activity jumpers. It nails the everyday — loveseat, stanley tumbler, lego star wars, instant pot, decorate a nursery, stainless steel cookware set, nintendo switch, all in motion — and brand queries redirect to curated brand pages. Attribute queries return the right product type but do not always honor the exact attribute.

How this was measured

- **Real shopper search, no production load.** Each term was run on Target's live search page and the top results read from what the browser rendered — bounded and paced, no crawling or scraping. Reading the raw search API at volume triggers bot protection, which we did not bypass.
- **Graded relevance.** Every returned product was scored 0–3 for relevance by a local large language model (Meta Llama 3.2 3B, temperature 0, reproducible), summarized as NDCG (0–1) and mean grade. 721 product judgments across 110 terms and five query types.
- **Why grade, not coverage.** A big-box engine returns something for almost anything, so coverage and NDCG (internal ordering) stay high; mean grade is the honest signal of whether results match intent.
- **Caveat.** Results were read from Target's organic search (sponsored and "recommended" carousel tiles excluded); the lowest-scoring terms were re-verified against the live site to rule out capture artifacts. A production run would use the full organic search API at scale.

Findings

From the 110-term evaluation, severity-ranked, with a fix candidate for each. Ranking these by demand is the next step (below).

Severity	Area	Finding
High	**Intent / job-function**	**Compositional natural-language queries return a plausible aisle, not the ask** Job-function queries grade 1.33/3 (63% not-relevant). move-into-a-new-house -¿ romance novels & toy food trucks; set-up-a-home-theater -¿ pajamas mixed with soundbars; furnish-a-living-room -¿ vases. Works only when the phrase maps to a single category (healthy snacks for kids, home gym equipment, organize my kitchen). Fix candidate: query understanding / intent expansion for multi-word task queries.

Severity	Area	Finding
Medium	Synonym equivalence	**Synonyms are substring-matched, not mapped — regional/UK variants fail** Variants underperform their US canonical: trainers 0.88 vs sneakers 1.88; rubbish bin 1.0 vs trash can 2.0; nappies 1.17 vs diapers 2.0; pram 1.63 vs stroller 2.5. Ambiguous tokens misfire: pop/popsicle -¿ Pop-Tarts, ice-lolly -¿ mints. crockpot=slow cooker and rc car=remote control car are handled perfectly. Fix candidate: a synonym dictionary (incl. en-GB) mapped to canonical categories.
Medium	Attribute exactness	**Attribute queries return the right product type but not always the exact attribute** A 65-inch QLED search surfaces mixed sizes/panel types; rose-gold-desk-lamp returns assorted desk lamps. The product type is right; the specific attribute (size/color/material) is loosely honored. Fix candidate: weight attribute facets in ranking.
Low	Brand navigation (strength)	**Brand-name queries correctly redirect to curated brand pages** threshold, cat & jack, good & gather, hearth & hand, all in motion redirect to curated brand pages (1,600-2,700 results). Caveat: the plp_search_v2 keyword API returns 0 for pure brand names, so they must be counted via the brand page (a measurement note, not a defect).
Info	Measurement	**Coverage is ˜100% — weight grade quality, not zero-result rate** Target answers nearly every query, so NDCG/coverage look strong; the mean grade (1.76/3) and not-relevant rate (38%) reveal the shallow answers. For big-box, rank fixes by grade-quality x demand, not by zero-result rate.

Demand-ranked quick wins

Failing terms (grade below 1.5 on demand-mappable query types) joined to Google Keyword Planner monthly search volume — the highest-volume failures are the highest-value fixes.

Fix candidate	Query type	Monthly volume	Grade
move into a new house	job-function	390	0.67
trainers	synonym	22,200	0.88
rubbish bin	synonym	480	1.00
furnish a living room	job-function	10	1.00
set up a home theater	job-function	390	1.00
start a garden	job-function	1,600	1.00
rose gold desk lamp	attribute	50	1.00
first apartment essentials	job-function	1,900	1.17
thanksgiving dinner table	job-function	1,600	1.17
outfit for a job interview	intent	14,800	1.17
organic cotton crib sheet	attribute	5,400	1.17
memory foam pillow	attribute	49,500	1.17
waterproof hiking boots	attribute	33,100	1.17
nappies	synonym	12,100	1.17
popsicle	synonym	201,000	1.17
cast iron skillet 12 inch	attribute	4,400	1.25
gift for a new mom	intent	14,800	1.25
back to school supplies	job-function	9,900	1.25
home office setup	job-function	2,900	1.33
warm coat for winter	intent	2,900	1.33
65 inch qled tv	attribute	1,600	1.33
sweater	synonym	246,000	1.33

Revenue at risk — and the opportunity at 1,000 terms

PROJECTED ACROSS THE TOP 1,000 SEARCH TERMS

$3.3M per year recoverable — range $853K to $8.3M. Extrapolated from the measured sample at $3,318 per term on average, across 83 demand-weighted terms.

Measured basis: across 83 search terms carrying real Google demand (of 110 evaluated), poor relevance puts $275K per year at risk today (range $71K–$688K); 53 of those terms carry loss. The 1,000-term figure applies the same per-term average to a retailer's highest-demand searches.

Model: for each searched term, monthly Google volume × Target market share (3.5%) × search conversion (2.5%) × average order value ($60) is the revenue at stake; the share forfeited scales with how poorly the term grades — a term grading 2/3 or better counts as no loss. 53 of 83 demand terms contribute.

Term	Monthly volume	Grade	Annual $ lost (base)
popsicle	201,000	1.17	$52,551
sweater	246,000	1.33	$51,918
jumper	110,000	1.50	$17,325
soda	165,000	1.67	$17,152
tennis shoes	135,000	1.63	$15,734
remote control car	135,000	1.67	$14,033
rc car	135,000	1.67	$14,033
memory foam pillow	49,500	1.17	$12,942

Modeled estimate for the sampled terms only — not audited revenue, and not extrapolated to Target's full query catalog (which is far larger). Tune market share, conversion and AOV to Target's real figures to tighten the range.

How we can help

COMPLIMENTARY NOW — DEEPER ON REQUEST

This is a complimentary read on a bounded query set. We can run the same evidence-first analysis **at scale** against your real on-site query logs — a **one-time** audit or a **recurring** check — and overlay Google search demand so the fixes are ranked by revenue impact. The deliverable: a reproducible, prioritized list of the exact searches to fix.

Relevance by query type

NDCG is the internal ordering (0–1); mean grade is the honest relevance (0–3); shallow = terms whose results average below grade 1.5.

Query type	Terms	NDCG	Mean grade	Not-relevant	Shallow
Synonym / category	36	88%	1.91	29%	5/36
Attribute (type+material)	25	95%	1.73	32%	6/25
Job-function (task)	18	90%	1.33	63%	11/18
Intent (natural language)	16	87%	1.60	47%	6/16
Literal (brand/product)	15	91%	2.07	34%	4/15

Where it breaks: shallowest searches

The lowest-grade searches — each returns results, but the wrong ones. "Top result" is the first product Target returned.

Search term	Query type	Grade	Top result
move into a new house	job_function	0.67	The Love Hypothesis - by Ali Hazelw
trainers	synonym	0.88	Munchkin Miracle 360 Trainer Cup 7oz - Blu
rubbish bin	synonym	1.00	2.5gal Waste Basket White - Brightroom : C
everything for a college dorm	job_function	1.00	Post-it Flags Assorted Bright Colors 100 F
host a backyard bbq	job_function	1.00	Drizzilicious Mini Rice Cakes Birthday Cak
furnish a living room	job_function	1.00	White Vase with Pink Bow Room Essentials
set up a home theater	job_function	1.00	Small Wood Bluetooth Radio - Hearth
start a garden	job_function	1.00	LEGO Botanicals Water Lilies Buildin
what to get a teenage girl	intent	1.00	Betting on You - by Lynn Painter (P
rose gold desk lamp	attribute	1.00	Task Lamp with Charging Pink - Room

Synonym equivalence: does each variant match its canonical?

Synonyms are grouped by the canonical term a shopper would expect them to match. Where a variant grades well below its canonical (a large negative Δ), the engine substring-matched the word rather than mapping the synonym — most misses are UK/regional variants ("trainers", "rubbish bin", "nappies") or ambiguous tokens ("pop" returns Pop-Tarts, not soda). Groups where every variant matches ("crockpot" = "slow cooker", "rc car" = "remote control car") are handled well.

Canonical (grade)	Variant	Grade	Δ	Variant top result
sneakers (1.88)	tennis shoes	1.63	-0.25	Kids Devyn Slip-On Performance
	trainers	0.88	-1.00	Munchkin Miracle 360 Trainer C
trash can (2.00)	garbage can	1.75	-0.25	2.5gal Waste Basket White - Br
	rubbish bin	1.00	-1.00	2.5gal Waste Basket White - Br
sofa (2.38)	couch	2.50	+0.12	Costway Floor Couch, Loveseat
	settee	1.50	-0.88	12pc Stoneware Westfield Dinne
	loveseat	2.88	+0.50	Dyonery 65" Folding Boneless C
stroller (2.50)	pram	1.63	-0.87	Baby Disney Winnie The Pooh Co

Canonical (grade)	Variant	Grade	Δ	Variant top result
	pushchair	2.25	-0.25	hauck Sport T13 Lightweight Co
diapers (2.00)	nappies	1.17	-0.83	The Honest Company Clean Consc
silverware (2.38)	flatware	2.50	+0.12	20pc Teagan Flatware Silver -
	cutlery	1.75	-0.63	Hefty Ecosave Compostable Cutl
comforter (2.13)	duvet	2.00	-0.13	King Heavyweight Linen Blend D
	quilt	2.25	+0.12	King Heavyweight Linen Blend Q
coffee table (2.25)	cocktail table	2.13	-0.12	Marble Drink Table - Threshold
slow cooker (2.17)	crockpot	2.17	0.00	Crock-Pot 4.5qt Slow Cooker Si
	crock pot	2.17	0.00	Crock-Pot 4.5qt Slow Cooker Si
soda (1.67)	pop	2.50	+0.83	Pop-Tarts Frosted Fruit Variet
	soft drinks	1.67	0.00	OLIPOP Shirley Temple Soda - 1
remote control car (1.67)	rc car	1.67	0.00	Sharper Image Toy RC Corvette
sweater (1.33)	jumper	1.50	+0.17	Baby Einstein Farm Explorers J
	pullover	1.75	+0.42	Womens Oversized Long Sleeve R
popsicle (1.17)	ice lolly	1.50	+0.33	Ice Cube Tray Soft Release wit

Appendix: all terms tried (with grade)

NDCG = internal ordering; Grade = mean relevance 0–3.

Search term	NDCG	Grade
Synonym / category		
trainers	100%	0.88
rubbish bin	66%	1.00
nappies	100%	1.17
popsicle	76%	1.17
sweater	74%	1.33
settee	80%	1.50
jumper	91%	1.50
ice lolly	85%	1.50
tennis shoes	94%	1.63
pram	98%	1.63
soda	100%	1.67
soft drinks	97%	1.67
remote control car	100%	1.67
rc car	100%	1.67
cutlery	78%	1.75
garbage can	69%	1.75
pullover	–	1.75
sneakers	94%	1.88
trash can	79%	2.00
duvet	100%	2.00
diapers	83%	2.00

Search term	NDCG	Grade
cocktail table	78%	2.13
comforter	100%	2.13
slow cooker	89%	2.17
crockpot	84%	2.17
crock pot	84%	2.17
coffee table	74%	2.25
pushchair	99%	2.25
quilt	94%	2.25
sofa	81%	2.38
silverware	84%	2.38
couch	81%	2.50
flatware	86%	2.50
stroller	97%	2.50
pop	96%	2.50
loveseat	100%	2.88
Attribute (type+material)		
rose gold desk lamp	100%	1.00
organic cotton crib sheet	100%	1.17
memory foam pillow	100%	1.17
waterproof hiking boots	100%	1.17
cast iron skillet 12 inch	66%	1.25
65 inch qled tv	88%	1.33
air fryer 8 quart	83%	1.50
marble coffee table	96%	1.67
55 inch 4k oled tv	–	1.71
walnut dining table	93%	1.75
high waisted black leggings	97%	1.83
blackout curtains 84 inch	98%	1.83
reusable stainless steel water bottle	98%	1.83
fleece lined leggings	98%	1.83
kids rolling backpack	98%	1.83
gaming chair with lumbar support	99%	1.83
standing desk adjustable	99%	1.83
velvet accent chair	99%	1.88
nonstick frying pan	99%	2.00
queen size cooling sheets	100%	2.00
wireless noise cancelling headphones	100%	2.00
bluetooth portable speaker	100%	2.00
stackable storage bins	100%	2.00
12 cup programmable coffee maker	80%	2.33
stainless steel cookware set	96%	2.67
Job-function (task)		
move into a new house	76%	0.67
everything for a college dorm	100%	1.00
host a backyard bbq	100%	1.00
furnish a living room	100%	1.00

Search term	NDCG	Grade
set up a home theater	100%	1.00
start a garden	76%	1.00
first apartment essentials	86%	1.17
thanksgiving dinner table	69%	1.17
back to school supplies	79%	1.25
home office setup	88%	1.33
kids birthday party supplies	96%	1.33
holiday party hosting	85%	1.50
newborn baby essentials	98%	1.50
camping trip gear	–	1.80
organize my closet	100%	1.83
home gym equipment	100%	2.00
small space storage solutions	100%	2.00
baby shower gifts	84%	2.17
Intent (natural language)		
what to get a teenage girl	100%	1.00
gift for a 5 year old boy	79%	1.17
outfit for a job interview	87%	1.17
reduce clutter in my bathroom	77%	1.17
gift for a new mom	81%	1.25
warm coat for winter	60%	1.33
cozy fall decor	90%	1.50
healthy snacks for kids	100%	1.67
gift for someone who loves coffee	75%	1.67
something fun for a road trip	92%	1.67
something to organize my kitchen	94%	1.75
keep drinks cold at the beach	100%	1.83
gift for a dog lover	79%	1.83
something to help me sleep better	79%	2.00
gift under 25 dollars	100%	2.00
decorate a nursery	95%	2.67
Literal (brand/product)		
bullseye dog toy	81%	1.17
good and gather	73%	1.33
room essentials	100%	1.33
figmint	73%	1.33
threshold	90%	1.50
dyson v8	69%	1.63
keurig k mini	100%	1.83
airpods pro	100%	2.00
cat and jack	95%	2.50
hearth and hand	95%	2.50
nintendo switch	100%	2.67
all in motion	95%	2.67
lego star wars	88%	2.83
instant pot	99%	2.83

Search term	NDCG	Grade
stanley tumbler	100%	2.88

Appendix B: revenue-loss calculation

Every dollar figure in this report is reproducible from the formula and parameters below, applied to real Google Keyword Planner volumes. It is a modeled estimate, not audited revenue.

Per term, per month: monthly_lost = volume × market_share × conversion_rate × AOV × loss_fraction, where loss_fraction = max(0, (accept_grade − grade)÷ accept_grade), with accept_grade = 2. Annual = monthly × 12; total = sum over demand-weighted terms.

Parameters (three scenarios):

Parameter	Conservative	Base	Aggressive
Market share	2.0%	3.5%	5.0%
Search conversion	1.5%	2.5%	3.5%
Average order value	$45	$60	$75

Per-term detail (base scenario, terms contributing loss):

Term	Vol / mo	Grade	Loss frac.	Annual $ (base)
popsicle	201,000	1.17	0.415	$52,551
sweater	246,000	1.33	0.335	$51,918
jumper	110,000	1.50	0.25	$17,325
soda	165,000	1.67	0.165	$17,152
tennis shoes	135,000	1.63	0.185	$15,734
remote control car	135,000	1.67	0.165	$14,033
rc car	135,000	1.67	0.165	$14,033
memory foam pillow	49,500	1.17	0.415	$12,942
sneakers	301,000	1.88	0.06	$11,378
waterproof hiking boots	33,100	1.17	0.415	$8,654
trainers	22,200	0.88	0.56	$7,832
soft drinks	49,500	1.67	0.165	$5,146
settee	27,100	1.50	0.25	$4,268
healthy snacks for kids	40,500	1.67	0.165	$4,210
outfit for a job interview	14,800	1.17	0.415	$3,869
gift for a new mom	14,800	1.25	0.375	$3,497
fleece lined leggings	60,500	1.83	0.085	$3,240
garbage can	40,500	1.75	0.125	$3,189
nappies	12,100	1.17	0.415	$3,164
marble coffee table	27,100	1.67	0.165	$2,817
cutlery	33,100	1.75	0.125	$2,607
pram	22,200	1.63	0.185	$2,587
back to school supplies	9,900	1.25	0.375	$2,339
pullover	22,200	1.75	0.125	$1,748
organic cotton crib sheet	5,400	1.17	0.415	$1,412

Term	Vol / mo	Grade	Loss frac.	Annual $ (base)
cast iron skillet 12 inch	4,400	1.25	0.375	$1,040
standing desk adjustable	18,100	1.83	0.085	$969
warm coat for winter	2,900	1.33	0.335	$612
home office setup	2,900	1.33	0.335	$612
walnut dining table	6,600	1.75	0.125	$520
start a garden	1,600	1.00	0.5	$504
first apartment essentials	1,900	1.17	0.415	$497
ice lolly	2,900	1.50	0.25	$457
thanksgiving dinner table	1,600	1.17	0.415	$418
reusable stainless steel water bottle	6,600	1.83	0.085	$353
65 inch qled tv	1,600	1.33	0.335	$338
gift for a dog lover	5,400	1.83	0.085	$289
air fryer 8 quart	1,300	1.50	0.25	$205
velvet accent chair	4,400	1.88	0.06	$166
move into a new house	390	0.67	0.665	$163
rubbish bin	480	1.00	0.5	$151
set up a home theater	390	1.00	0.5	$123
cozy fall decor	480	1.50	0.25	$76
high waisted black leggings	1,000	1.83	0.085	$54
gaming chair with lumbar support	1,000	1.83	0.085	$54
55 inch 4k oled tv	390	1.71	0.145	$36
gift for someone who loves coffee	320	1.67	0.165	$33
blackout curtains 84 inch	390	1.83	0.085	$21
rose gold desk lamp	50	1.00	0.5	$16
holiday party hosting	90	1.50	0.25	$14
organize my closet	170	1.83	0.085	$9
camping trip gear	90	1.80	0.1	$6
furnish a living room	10	1.00	0.5	$3
Total (base)				**$275,384**

Extrapolation to 1,000 terms: $275,384 ÷ 83 terms = $3,318 per term (base); × 1,000 = **$3.3M per year** (conservative $853K, aggressive $8.3M).

Illustrative projection: assumes the next 1000 search terms share a similar average demand and failure profile as the measured sample. Real query catalogs follow a long-tail demand distribution, so validate against actual query logs.